

Name: _____

R. Hammack

Score: _____

Directions: Put away phones, calculators, smart watches, etc. Answer in the space provided.

1. (4 points) Consider the vectors $\mathbf{u} = (8, -2, 0, 1)$, $\mathbf{v} = (3, -2, 1, 1)$ and $\mathbf{w} = (1, 4, 4, 1)$. in $\mathbb{R}^4$. Find $\mathbf{x}$, given that $\mathbf{w} - 2\mathbf{x} = \mathbf{u} - 3\mathbf{v}$.

$$\begin{aligned}
 \mathbf{w} - 2\mathbf{x} &= \mathbf{u} - 3\mathbf{v} \\
 -2\mathbf{x} &= \mathbf{u} - 3\mathbf{v} - \mathbf{w} \\
 \mathbf{x} &= -\frac{1}{2}(\mathbf{u} - 3\mathbf{v} - \mathbf{w}) \\
 \mathbf{x} &= -\frac{1}{2}[(8, -2, 0, 1) - 3(3, -2, 1, 1) - (1, 4, 4, 1)] \\
 \mathbf{x} &= -\frac{1}{2}(-2, 0, -7, -3) \\
 \mathbf{x} &= \left(1, 0, \frac{7}{2}, \frac{3}{2}\right)
 \end{aligned}$$

2. (8 points) Say if each of the following sets is a vector space. (If so, just answer “Yes.” If it is **not** a vector space, then answer “No” and state a vector space property that fails.)

- (a) $\{(x, y) \in \mathbb{R}^2 : x - y = 0\}$ YES
- (b) $\left\{\begin{bmatrix} x & x+1 \\ x & x \end{bmatrix} : x \in \mathbb{R}\right\}$ NO, because there is no zero vector $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.
- (c) $\left\{\begin{bmatrix} 0 & x \\ 2x & 0 \end{bmatrix} : x \in \mathbb{R}\right\}$ YES
- (d) P_9 YES
- (e) $M_{9,3}$ YES
- (f) $\{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$ NO, because is not a negative of every vector
- (g) $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 > 0\}$... NO. This set does not contain the zero vector $(0, 0)$.
- (h) $\{(x, y, z) \in \mathbb{R}^3 : x = 0\}$ YES